Fixed Point Properties of Group Actions on Trees

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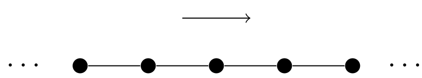
Definition 1 (Tree)**.** A *tree* is a graph $T = (V, E)$ that is connected and has no cycles.

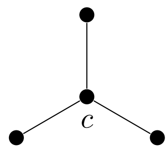
Definition 2 (Group action on a tree)**.** An *action of a group G on a tree T* is a group homomorphism $G \longrightarrow \mathrm{Aut}(T)$.

Example 3. Here are two examples of group actions on trees:

(1) The group $\mathbb{Z}$ acts without fixed points by translation on the infinite line graph:

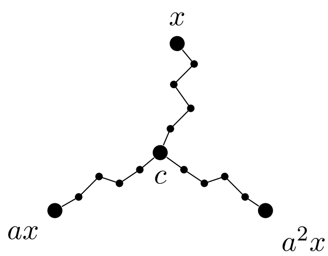


(2) The group $\mathbb{Z}/3\mathbb{Z}$ acts by rotation on the following tree, fixing the central vertex c:



Proposition 4. *Every action of $\mathbb{Z}/3\mathbb{Z} = \langle a \rangle$ on a tree has a fixed point.*

Proof. Pick an arbitrary vertex x and consider its orbit $\{x, ax, a^2x\}$ under $\langle a \rangle$. In a tree, there is a unique geodesic between any two vertices. Let $[x, ax]$ be the unique geodesic between x and ax. Since there are no cycles in a tree, the intersection of $[x, ax]$, $[ax, a^2x]$, and $[a^2x, x]$ is a single point c. Since a permutes these three vertices, it must fix c.



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Remark 5. *In fact, every finite group must act with fixed points on every tree. The proof is similar: one can always find a center of the finite orbit of a single point. In the case of $\mathbb{Z}/3\mathbb{Z}$, the center is the vertex c.*

Question 6. *Which groups must act with a fixed point on every tree?*

Proposition 7. *$SL_2(\mathbb{Z})$ can act on a tree without fixed points.*

We now construct an action of $SL_2(\mathbb{Z})$ on a tree without fixed points.

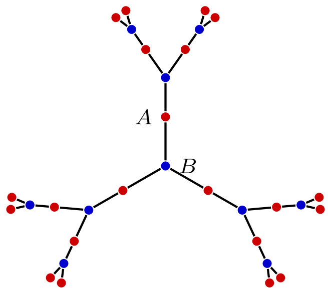
Construction 8. *Recall that $SL_2(\mathbb{Z})$ has the following representation:*

$$SL_2(\mathbb{Z}) \cong \langle x, y \mid x^4 = I,\ y^6 = I,\ x^2 = y^3 \rangle,$$

where

$$x = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad y = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}.$$

We first construct an infinite tree T whose vertices have degree 2 or 3, with the following patterns: every degree-3 vertex is adjacent to three degree-2 vertices, and every degree-2 vertex is adjacent to two degree-3 vertices.



Now we construct the action of $SL_2(\mathbb{Z})$ on T. The generators act as rotations on this tree:

- *x acts by a $180°$ rotation fixing only the vertex A;*
- *y acts by a $120°$ rotation fixing only the vertex B.*

Since x fixes only A and y fixes only B, this action has no global fixed point.

But what essentially makes $SL_2(\mathbb{Z})$ admit a fixed-point-free action on a tree? In fact, this construction has a natural algebraic explanation. The above representation of $SL_2(\mathbb{Z})$ reveals that $SL_2(\mathbb{Z})$ is obtained by gluing two subgroups $\langle x \rangle \cong \mathbb{Z}/4\mathbb{Z}$ and $\langle y \rangle \cong \mathbb{Z}/6\mathbb{Z}$ together through a common subgroup $\langle x^2 \rangle \cong \langle y^3 \rangle \cong \mathbb{Z}/2\mathbb{Z}$. Indeed, such structure is called an *amalgam*, and is denoted by

$$SL_2(\mathbb{Z}) \cong \mathbb{Z}/4\mathbb{Z} \ *_{\mathbb{Z}/2\mathbb{Z}} \ \mathbb{Z}/6\mathbb{Z}.$$

In fact, the structure of an amalgam is closely related to the existence of a fixed-point-free action on a tree.

Fact 9. *Every amalgamated product $G_1 *_H G_2$ with H properly embeded in both G_1 and G_2 can act on a tree without global fixed points.*

The construction is similar to that of $SL_2(\mathbb{Z})$.